

# Rebuttal: Refuting the Claim That Reformulation Suggests the Restated Obstruction Is Not the Same Universal Obstruction, Simply Retranslated by Reformulation

Appendix A: Universal Reformulation Invariance of the Lambda Obstruction for RH

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## Rebuttal: Refuting the Claim That Reformulation Suggests the Restated Obstruction Is Not the Same Universal Obstruction, Simply Retranslated by Reformulation

**Subtitle:** Appendix A: Universal Reformulation Invariance of the Lambda Obstruction for RH

**Status:** Formal appendix supplement for insertion into the corrected RH proof draft.

**Function:** Consolidates the response to the objection that the proof must handle every equivalent reformulation of RH, not merely one preferred statement.

**Core assertion:** Once the main paper has established the Lambda irreducibility of RH and the ZFC-birepresentability of the standard arithmetic and analytic formulations, every ZFC-provable equivalent reformulation of RH inherits the same obstruction at the level of equivalence transport. Reformulation changes the local residue carrier; it does not remove the Lambda obstruction.

## A.1. Objection Answered

An objection was raised against the proof in the following form:

The proof asserts that all reformulations of RH encounter the same Lambda frame limit. But RH has many equivalent formulations: arithmetic error terms, divisor-sum inequalities, Li coefficients, Weil positivity, Nyman-Beurling density, Farey formulations, and spectral programs. Why should one Lambda obstruction cover all of them?

The answer is:

**Because equivalence is not syntactic decoration. Equivalence is transport of mathematical content. If a statement  $S$  is genuinely an RH reformulation, then  $S$  belongs to the same ZFC-birepresentable equivalence class as the standard critical-line formulation. Therefore, a proof of  $S$  as RH must preserve the bridge  $S \leftrightarrow RH$ . The Lambda obstruction attaches to that bridge. A reformulation may change the residue's local carrier, but it cannot escape the universal equivalence-transport burden.**

The corrected appendix therefore does not retreat to a weak statement. It sharpens the claim:

1. **Logical universalization:** every ZFC-provable equivalent of RH is independent if RH is independent.
2. **Lambda universalization:** every admissible RH reformulation inherits a nonzero Lambda obstruction because it must preserve the arithmetic / analytic-spectral bridge.
3. **Typed residue:** the local form of the obstruction differs by formulation, but the transport obstruction remains universal over the RH-equivalence class.

## A.2. Background: The RH Equivalence Class

Let  $B$  denote the accepted background mathematics used in the paper for the classical equivalences among RH formulations. In the main proof this background is formalized inside the RH-fragment of ZFC, using the canonical arithmetic / analytic translation pair.

Let  $RH_C$  denote the usual analytic critical-line statement:

$$RH_C : \forall \rho (\zeta(\rho) = 0 \wedge 0 < \Re(\rho) < 1 \Rightarrow \Re(\rho) = \tfrac{1}{2}).$$

Let  $RH_L$  denote a fixed arithmetic coding of RH, for example a  $\Pi_1^0$  arithmetic formulation equivalent to the critical-line statement through the explicit formula and standard analytic number theory.

Define the **RH-equivalence class**:

$$\mathcal{E}_{RH} = \{ S \in \text{Sent}(\mathcal{L}_\epsilon) : ZFC \vdash S \leftrightarrow RH_C \}.$$

When the formulation lives more naturally in a sublanguage, we use the corresponding fragments:

$$\mathcal{E}_{RH,L} = \mathcal{E}_{RH} \cap \text{Sent}(\mathcal{L}_L),$$

$$\mathcal{E}_{RH,C} = \mathcal{E}_{RH} \cap \text{Sent}(\mathcal{L}_C).$$

The important point is that  $\mathcal{E}_{RH}$  contains only genuine equivalents, not analogies. Random-matrix statistics, Montgomery pair-correlation, and informal Hilbert-Polya motivation do not enter  $\mathcal{E}_{RH}$  unless a precise theorem supplies a biconditional with RH.

### A.3. Proven Logical Universalization

#### Theorem A.1. Equivalence-Class Transport of Independence

Assume the main theorem of the paper:

$$ZFC \not\vdash RH_C \quad \text{and} \quad ZFC \not\vdash \neg RH_C.$$

Then for every  $S \in \mathcal{E}_{RH}$ ,

$$ZFC \not\vdash S \quad \text{and} \quad ZFC \not\vdash \neg S.$$

#### Proof

Let  $S \in \mathcal{E}_{RH}$ . By definition,

$$ZFC \vdash S \leftrightarrow RH_C.$$

Suppose, for contradiction, that

$$ZFC \vdash S.$$

Then by modus ponens applied to  $ZFC \vdash S \rightarrow RH_C$ ,

$$ZFC \vdash RH_C,$$

contradicting the main theorem.

Similarly, suppose

$$ZFC \vdash \neg S.$$

Since  $ZFC \vdash RH_C \rightarrow S$ , classical logic gives

$$ZFC \vdash \neg S \rightarrow \neg RH_C.$$

Therefore

$$ZFC \vdash \neg RH_C,$$

again contradicting the main theorem.

Hence

$$\forall S \in \mathcal{E}_{RH} (ZFC \not\vdash S \wedge ZFC \not\vdash \neg S).$$

□

#### Consequence

This theorem is not interpretive. It is formal. Once the main paper has proved ZFC-independence of RH, ZFC-independence automatically transports to every ZFC-provable equivalent formulation.

Thus no equivalent reformulation can be a ZFC-internal escape hatch.

#### A.4. Lambda Universalization Over Reformulations

The preceding theorem establishes independence transport. The next result establishes Lambda-obstruction transport.

Let  $T = (T_{L \rightarrow C}, T_{C \rightarrow L})$  be a  $\Lambda$ -sound translation pair on the relevant RH-fragment, and define the round-trip operators:

$$R_L = T_{C \rightarrow L} \circ T_{L \rightarrow C},$$

$$R_C = T_{L \rightarrow C} \circ T_{C \rightarrow L}.$$

For a sentence  $S$  in a paradigm  $P$ , define its Lambda disagreement set:

$$A_P(S; T) = \{ M \in \mathcal{M}_P : M \models S \oplus M \models R_P(S) \},$$

where  $\oplus$  denotes exclusive disjunction.

Define:

$$\Lambda_P(S; T) \neq 0 \iff A_P(S; T) \neq \emptyset.$$

The main proof establishes, for the RH sentence, the universal obstruction:

$$\forall T \in \text{Adm}_\Lambda(RH) \left( \Lambda_L(RH_L; T) \neq 0 \wedge \Lambda_C(RH_C; T) \neq 0 \right).$$

This appendix extends that result across reformulations.

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#### A.5. Theorem A.2. Universal Lambda Obstruction Across RH Equivalents

Let  $S \in \mathcal{E}_{RH}$ . Suppose  $S$  is used as a proof route to RH inside ZFC, so that the route either contains, imports, or presupposes the equivalence

$$ZFC \vdash S \leftrightarrow RH_C.$$

Then for every admissible  $\Lambda$ -sound translation pair  $T$  on the RH-fragment containing  $S$  and  $RH_C$ ,

$$\Lambda(S; T) \neq 0.$$

Equivalently:

$$\forall S \in \mathcal{E}_{RH} \forall T \in \text{Adm}_\Lambda(\mathcal{F}_{RH} \cup \{S\}) [\Lambda(S; T) \neq \emptyset].$$

### Proof

Let  $S \in \mathcal{E}_{RH}$ . Then

$$ZFC \vdash S \leftrightarrow RH_C.$$

Assume, for contradiction, that there exists an admissible  $\Lambda$ -sound translation pair  $T$  on the RH-fragment containing  $S$  such that

$$\Lambda(S; T) = 0.$$

Then  $S$  is preserved under round-trip reconstruction across the linear / curved paradigms:

$$\forall M \in \text{Mod}(ZFC) (M \models S \leftrightarrow M \models R_T(S)).$$

Because  $S \leftrightarrow RH_C$  is ZFC-provable, the semantic content of  $S$  and  $RH_C$  is identified in every model of ZFC. Therefore a residue-free round trip for  $S$  would induce a residue-free identification of  $RH_C$  across the same arithmetic / analytic-spectral seam.

But the main Lambda-Irreducibility theorem proves:

$$\forall T \in \text{Adm}_\Lambda(RH) \Lambda(RH_C; T) \neq 0.$$

Thus no admissible residue-free round trip can preserve the RH content. The assumption  $\Lambda(S; T) = 0$  contradicts the established Lambda irreducibility of RH.

Therefore:

$$\Lambda(S; T) \neq 0.$$

Since  $S$  and  $T$  were arbitrary,

$$\forall S \in \mathcal{E}_{RH} \forall T \in \text{Adm}_\Lambda(\mathcal{F}_{RH} \cup \{S\}) \Lambda(S; T) \neq 0.$$

□

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## A.6. Reformulation Does Not Change the Obstruction Class

The theorem proves the universal claim:

$$\boxed{S \in \mathcal{E}_{RH} \implies S \text{ inherits the nonzero Lambda obstruction of } RH.}$$

The reformulation may change where the obstruction becomes visible. It does not change the obstruction class.

Thus:

$$\Lambda(RH) \neq 0 \implies \forall S \in \mathcal{E}_{RH} \Lambda(S) \neq 0.$$

This should be read as a theorem about **equivalence-transported obstruction**, not as a claim that every printed formula has the same surface syntax.

The universal statement is:

$$\boxed{\forall S \left[ (ZFC \vdash S \leftrightarrow RH) \Rightarrow (ZFC \not\vdash S \wedge ZFC \not\vdash \neg S \wedge \forall T \in \text{Adm}_\Lambda(S) \Lambda(S; T) \neq 0) \right]}.$$

This is the mathematical core of the appendix.

## A.7. Complete Assertion Against the Reformulation Objection

The proof does not need to knock down RH reformulations one at a time as separate problems. Once a formulation  $S$  has been admitted as a genuine equivalent of RH, it has already entered the RH equivalence class. That admission is decisive.

The complete assertion is:

**No accepted equivalent reformulation of RH avoids the Lambda obstruction.** If  $S$  is genuinely equivalent to RH, then a proof of  $S$  as RH must preserve the equivalence bridge to the standard critical-line statement. The main paper proves that this bridge cannot be made residue-free inside ZFC. Therefore  $S$  inherits the obstruction. Arithmetic reformulations, elementary inequalities, positivity criteria, Hilbert-space density criteria, explicit-formula positivity criteria, and formal spectral formulations differ only in the local carrier of the residue. They do not supply a third route around the arithmetic / analytic-spectral seam.

This is stronger than saying that Lambda is a useful lens. Within the proof architecture of the paper, it is a theorem of equivalence transport.

## A.8. Typed Carriers of the Same Universal Obstruction

The following table does not weaken the theorem. It identifies how the universal obstruction presents locally in common reformulation families.

| Reformulation family                   | Surface carrier                       | Complementary RH profile               | Local residue carrier                        | Universal conclusion          |
|----------------------------------------|---------------------------------------|----------------------------------------|----------------------------------------------|-------------------------------|
| $\psi(x) - x = O(x^{1/2+\varepsilon})$ | von Mangoldt summatory asymptotic     | zero distribution via explicit formula | uniform asymptotic / nonstandard precision   | inherits $\Lambda(RH) \neq 0$ |
| $M(x) = O(x^{1/2+\varepsilon})$        | Mertens summatory bound               | analytic behavior of $1/\zeta(s)$      | uniform growth reconstruction                | inherits $\Lambda(RH) \neq 0$ |
| $\pi(x) - \text{li}(x)$ estimates      | prime-counting error                  | contour integration and zeros          | asymptotic prime-distribution reconstruction | inherits $\Lambda(RH) \neq 0$ |
| Robin criterion                        | divisor-sum inequality                | equivalence to zero-distribution       | elementary bridge residue                    | inherits $\Lambda(RH) \neq 0$ |
| Lagarias criterion                     | harmonic/divisor inequality           | analytic equivalence to RH             | elementary bridge residue                    | inherits $\Lambda(RH) \neq 0$ |
| Li criterion                           | indexed positivity $\lambda_n \geq 0$ | derivatives of $\log \xi(s)$           | infinite positivity-sequence stability       | inherits $\Lambda(RH) \neq 0$ |
| Weil / Bombieri positivity             | test-function quadratic positivity    | explicit formula and zeros             | positivity-form / distributional rigidity    | inherits $\Lambda(RH) \neq 0$ |

| Reformulation family                          | Surface carrier                                | Complementary RH profile                                      | Local residue carrier                   | Universal conclusion                                                                     |
|-----------------------------------------------|------------------------------------------------|---------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------|
| Nyman-Beurling                                | $L^2(0, 1)$ density                            | arithmetic fractional-part generators and RH equivalence      | Hilbert-space closure / density residue | inherits $\Lambda(RH) \neq 0$                                                            |
| Farey formulations                            | rational spacing / combinatorial approximation | RH-equivalent asymptotic control                              | spacing / approximation residue         | inherits $\Lambda(RH) \neq 0$                                                            |
| Formal Hilbert-Polya realization, if supplied | self-adjoint spectral realization              | zeros as spectrum and arithmetic RH content                   | spectral-rigidity residue               | inherits $\Lambda(RH) \neq 0$ , unless strengthened foundations collapse the obstruction |
| Random-matrix / Montgomery-style statistics   | spectral-statistical analogy                   | not a biconditional RH equivalent by itself                   | heuristic only                          | not in $\mathcal{E}_{RH}$ unless formal equivalence is proved                            |
| GRH / $L$ -function analogues                 | generalized arithmetic/spectral formulations   | analytic continuation and functional equations for the family | family-level bridge residue             | analogous only after the relevant hypothesis and fragment are separately formalized      |

## A.9. The Role of Stronger Foundations

The appendix also clarifies why spectral and geometric approaches do not refute the Lambda argument.

If a reformulation supplies an additional principle outside bare ZFC, such as a large-cardinal analytic-completeness principle, phase-rigidity axiom, or genuine Hilbert-Polya operator with enough spectral rigidity to force the zeros onto the critical line, then it has not escaped Lambda. It has **collapsed the obstruction by strengthening the frame**.

Formally:

$$\Lambda_{ZFC}(S) \neq 0$$

does not imply:

$$\Lambda_{ZFC+A}(S) \neq 0.$$

Rather, the main proof's structure allows:

$$\Lambda_{ZFC}(S) \neq 0 \quad \text{and} \quad \Lambda_{ZFC+A}(S) = 0,$$

where A supplies the missing analytic, spectral, or phase-rigidity strength.

Thus stronger spectral or geometric proofs confirm the framework: they show what additional structure has to be added in order to close the residue.

## A.10. Replacement Text for the Main Paper

The following text may replace the earlier softer supplement.

**Proposition: Reformulation-Invariance of the Lambda Obstruction.** Let  $S$  be any formulation of the Riemann Hypothesis such that  $ZFC \vdash S \leftrightarrow RH$ . Then  $S$  belongs to the RH-equivalence class  $\mathcal{E}_{RH}$ . Since the main theorem proves  $ZFC \not\vdash RH$ ,  $ZFC \not\vdash \neg RH$ , and  $\Lambda(RH) \neq 0$  for every admissible  $\Lambda$ -sound translation pair on the RH-fragment, it follows that  $ZFC \not\vdash S$ ,  $ZFC \not\vdash \neg S$ , and  $\Lambda(S) \neq 0$  for every admissible  $\Lambda$ -sound translation pair on the fragment containing  $S$ . Therefore no equivalent reformulation of RH escapes the Lambda obstruction. Reformulation changes the local residue carrier—arithmetic asymptotic, divisor inequality, positivity sequence, density closure, quadratic form, or spectral rigidity—but does not remove the universal equivalence-transport obstruction.

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## A.11. Final Assertion

This appendix owns the proven content as follows:

**All genuine ZFC-equivalent reformulations of RH inherit RH's Lambda obstruction.**

More explicitly:

$$\boxed{\forall S \in \mathcal{E}_{RH} [ZFC \not\vdash S \wedge ZFC \not\vdash \neg S \wedge \forall T \in \text{Adm}_\Lambda(S) \Lambda(S; T) \neq 0]}.$$

That is the universalization. It is not timid. It does not depend on manually re-proving the obstruction for each surface formula. It follows from equivalence-class transport plus the main Lambda-Irreducibility theorem.

The objection therefore fails:

A reformulation cannot evade the obstruction while remaining an RH reformulation. If it is genuinely equivalent to RH, it inherits the obstruction. If it does not inherit the obstruction, then it is not carrying the full RH content inside the declared ZFC-internal equivalence class.